

Shashank Gupta

shashankgupta496@gmail.com | +91-6266806450 | Bhopal, India | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student with practical experience building AI-powered applications using Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), and multi-agent AI pipelines. Proficient in Python and prompt engineering with hands-on project experience in FAISS vector stores, Hugging Face models, and OpenAI API. Passionate about developing intelligent systems and eager to contribute to real-world AI products in an early-stage startup environment.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript

AI / LLM Frameworks: LangChain, LangGraph (familiar), OpenAI API, Hugging Face Inference API

RAG & Vector Search: FAISS, sentence-transformers, RecursiveCharacterTextSplitter, ConversationalRetrievalChain

LLM Models: Llama 3.1-8B-Instruct, GPT-4, Claude, MiniLM (all-MiniLM-L6-v2)

Prompt Engineering: Multi-stage agent prompting, schema-constrained generation, prompt injection protection

Web Frameworks: Next.js, Express.js, Streamlit

Developer Tools: Git, GitHub, Node.js, npm, pip, REST APIs

AI Concepts: AI Agents, RAG Systems, LLM Orchestration, Vector Embeddings, Knowledge Bases

Other: FAISS (CPU), PyPDF2, python-docx, Speech Recognition API, Vercel, Render

PROJECTS

LangChain RAG Chat — Voice-Enabled RAG Chatbot

2026

Python | LangChain | Hugging Face | FAISS | Llama 3.1 | Streamlit | GitHub:

github.com/ShashankGupta9/Langchain-RAG-CHAT

- Built a production-ready Retrieval-Augmented Generation (RAG) chatbot using LangChain's ConversationalRetrievalChain backed by Meta's Llama 3.1-8B-Instruct model via Hugging Face Inference API.
- Implemented document ingestion pipeline supporting PDF, DOCX, and TXT files — splitting content into overlapping chunks (1000 tokens, 200 overlap) using RecursiveCharacterTextSplitter and indexing with FAISS vector store.
- Generated semantic embeddings using sentence-transformers (all-MiniLM-L6-v2) for accurate similarity-based document retrieval, enabling context-aware, grounded AI responses.
- Developed a custom Streamlit voice component using browser Speech Recognition API (STT) and Speech Synthesis API (TTS), enabling fully hands-free interaction with the AI system.
- Designed multi-session chat history management, dark/light theme UI, and secure API key handling — demonstrating full-stack AI product development skills.

AI Deterministic UI Generator — Multi-Agent LLM Pipeline

2026

Node.js | Express.js | Next.js | OpenAI API | Prompt Engineering | GitHub: github.com/ShashankGupta9/AI-UI_Generator

- Designed and built a full-stack AI application that converts natural language prompts into structured, deterministic, production-ready UI components using a multi-stage AI agent pipeline.
- Engineered a 4-stage agentic workflow — Planner → Generator → Validator → Explainer — where each stage uses targeted prompt engineering to produce reliable, schema-constrained JSON outputs.
- Applied advanced prompt engineering techniques including component whitelisting, schema validation, and prompt injection protection to ensure safe and consistent LLM-generated outputs.

- Built a real-time split-panel interface (chat + live preview) in Next.js and deployed the system with Express.js backend on Render and frontend on Vercel.
- Demonstrated ability to evaluate AI output quality, iterate on prompts, and build robust AI systems — core responsibilities aligned with the AI Engineering Intern role.

Seva-Kart — Full-Stack Service Marketplace

2025

TypeScript | Next.js | REST APIs | GitHub: github.com/ShashankGupta9/Seva-Kart

- Built a full-stack web platform connecting users with local service providers (plumbers, electricians, cleaners) with service discovery, booking, and management features.
- Demonstrates product development skills — end-to-end feature ownership, API design, and frontend/backend integration — applicable to building AI-integrated product features at a startup.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering

Expected 2027

Sanjeev Agrawal Global Education (SAGE), Bhopal, India | CGPA: 7.5 / 10

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Software Engineering

KEY STRENGTHS

- Hands-on experience with LangChain, RAG systems, FAISS vector stores, and Hugging Face — directly matching PrepOwl's AI Engineering Intern requirements.
- Proven ability to build, test, and deploy AI agents and LLM-powered applications from scratch with zero formal guidance.
- Strong prompt engineering skills demonstrated through multi-stage agent design, schema-constrained generation, and output validation.
- Self-driven learner with a problem-solving mindset — comfortable working in fast-paced, early-stage startup environments.
- Familiar with ChatGPT, Claude, and Gemini — actively uses state-of-the-art LLMs for development and experimentation.